

Retro Bowl

Art & Sprite Specification

Read out of the shipping build, 14 September 2026 · prepared for Retro Buhurt (BonkWorks)

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Every number in this document was parsed out of Retro Bowl's own asset manifest — `RetroBowl.js` (3.0 MB) and its two texture atlases — not read off a community guide. That gives exact canvas sizes, frame counts, playback speeds and draw origins for all **205 sprites and 632 frames**, the five fonts by name, and the full source of the shader that puts a team's colours on a player.

Where this document says a thing *is* a size, that size came out of the manifest. Where it reads intent into a decision, it says so. The last section maps all of it onto Retro Buhurt as the code stands today, and lists the four places the two designs currently disagree.

1 The canvas

Retro Bowl is a **480 × 270** game. Thirty-one of its thirty-three rooms are exactly that size, and so are their cameras. 480 × 270 is 1080p divided by four, which is why it scales to any modern screen with no filtering and no half-pixels.

ROOM	SIZE	CAMERA	WHAT IT IS
31 rooms	480×270	480×270	every menu, roster, league, draft, hall of fame, options screen
rm_match	2600×540	960×540	the only room you play in
rm_kongregate	960×540	—	portal splash, not gameplay

The match room is not a second resolution. It is the same 480 × 270 art grid in doubled coordinates. The field art is 1300 px wide and fills a 2600-wide room, so it is drawn at ×2; the camera is 960 × 540, so you are looking at **480 art-pixels of field at a time** — the same density as every menu. Working in 2× space buys sub-pixel movement for twenty-two running men without ever showing a half-pixel. The art never leaves the 480 × 270 grid.

The vertical budget is exact and worth copying: the field sprite is 237 px tall and the crowd strip above it is 33 px. 237 + 33 = 270. Nothing is left over and nothing is cropped.

The 1300×237 field, and the 480×270 window onto it

spr_fieldfans — 1300×33, 2 frames

spr_field — 1300×237, 7 variants

camera: 480 art-px wide
1300 px — the whole pitch, end zone to end zone

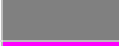
The camera pans along the length. Endzones (spr_endzone, 100×169) and goal posts (spr_goal_post, 17×90) are drawn OVER the field, not painted into it - they carry club colour.

Screen geometry, to scale.

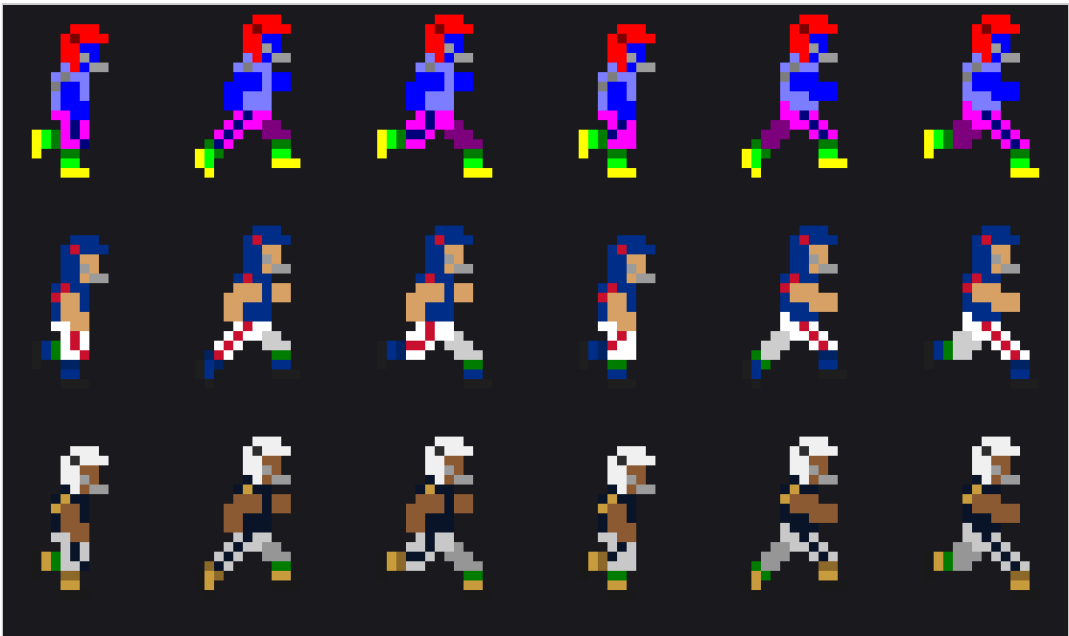
2 The palette system — the load-bearing decision

This is the part of Retro Bowl's art that is not obvious from playing it, and it is the part worth stealing. **The player sprites are not coloured artwork. They are masks.**

Every on-field figure is painted in eleven flat, arbitrary key colours — pure red, pure magenta, pure blue and so on. Nothing in the file is the colour you see in the game. At draw time a fragment shader, `sh_ColorReplaceBlend`, matches each pixel against the eleven keys and substitutes that team's kit colour for that slot. One set of sprites serves all thirty-two clubs, four kits each, home and away, for ever.

SHADER UNIFORM	KEY	WHAT IT PAINTS
colorHelmet		#FF0000 helmet shell
colorLogo		#800000 helmet stripe / logo
colorShirt		#8080FF jersey body
colorSleeve		#808080 jersey sleeve
colorPants		#FF00FF pants
colorPants_b		#800080 pants shade
colorStripe		#000080 pant stripe
colorSocks		#00FF00 socks upper
colorSocks_b		#008000 socks lower
colorSkin		#0000FF skin — arms, legs, face
colorCleats		#FFFF00 cleats

The eleven key colours, exactly as the shader declares them. Every one is a full-bright or half-bright primary — which is how the artist keeps them straight while drawing, and how the comparison stays exact.



The six-frame run cycle. Top: as it is stored in the atlas. Middle and bottom: the same six frames through a re-implementation of their shader with two different kits. One drawing, any club.

The shader, in full

```
// sh_ColorReplaceBlend - fragment shader, verbatim from the build
uniform vec3 colorHelmet, colorLogo, colorShirt, colorSleeve, colorPants,
            colorPants_b, colorStripe, colorSocks, colorSocks_b,
            colorSkin, colorCleats;           // the eleven KEYS
```

```
uniform vec3 replaceHelmet, replaceLogo, ... replaceCleats; // this club's kit

void main() {
    vec4 pixel = texture2D( gm_BaseTexture, v_vTexcoord );
    if (pixel.a == 1.0) {
        vec3 eps = vec3(0.009, 0.009, 0.009);
        if      (within(pixel, colorShirt,  eps)) pixel = vec4(replaceShirt, 1.0);
        else if (within(pixel, colorSkin,   eps)) pixel = vec4(replaceSkin, 1.0);
        else if (within(pixel, colorHelmet, eps)) pixel = vec4(replaceHelmet, 1.0);
        ... nine more, in a fixed priority order ...
    }
    gl_FragColor = pixel;
}
```

Three consequences the artist has to know before drawing anything.

- **Only fully opaque pixels are recoloured.** The shader tests `pixel.a == 1.0`. Any anti-aliased edge, any soft shadow, any 50%-alpha pixel *keeps its key colour* and renders as raw magenta on screen. This is why Retro Bowl has no anti-aliasing anywhere: the palette system forbids it.
- **The match is exact to ± 0.009** (about 2/255). A key colour that is one step off is not recoloured. Colours must be typed in, never eyedropped, never resampled, never saved through a lossy step.
- **Shading is a second slot, not a darker version of the first.** Pants and pants-shade are two separate uniforms; so are socks and socks-shade. The artist gets two tones per garment and the club decides what both of them are. There is no automatic darkening.

What is not a key colour

Black, white, the facemask grey, and the brown of the ball are fixed — they are drawn once and never swapped. Notably there is **no outline colour at all**. Retro Bowl's figures have no black keyline; they read against the field purely on value contrast between the kit and the grass, which is the reason the kit palettes are as saturated as they are.

The kit data

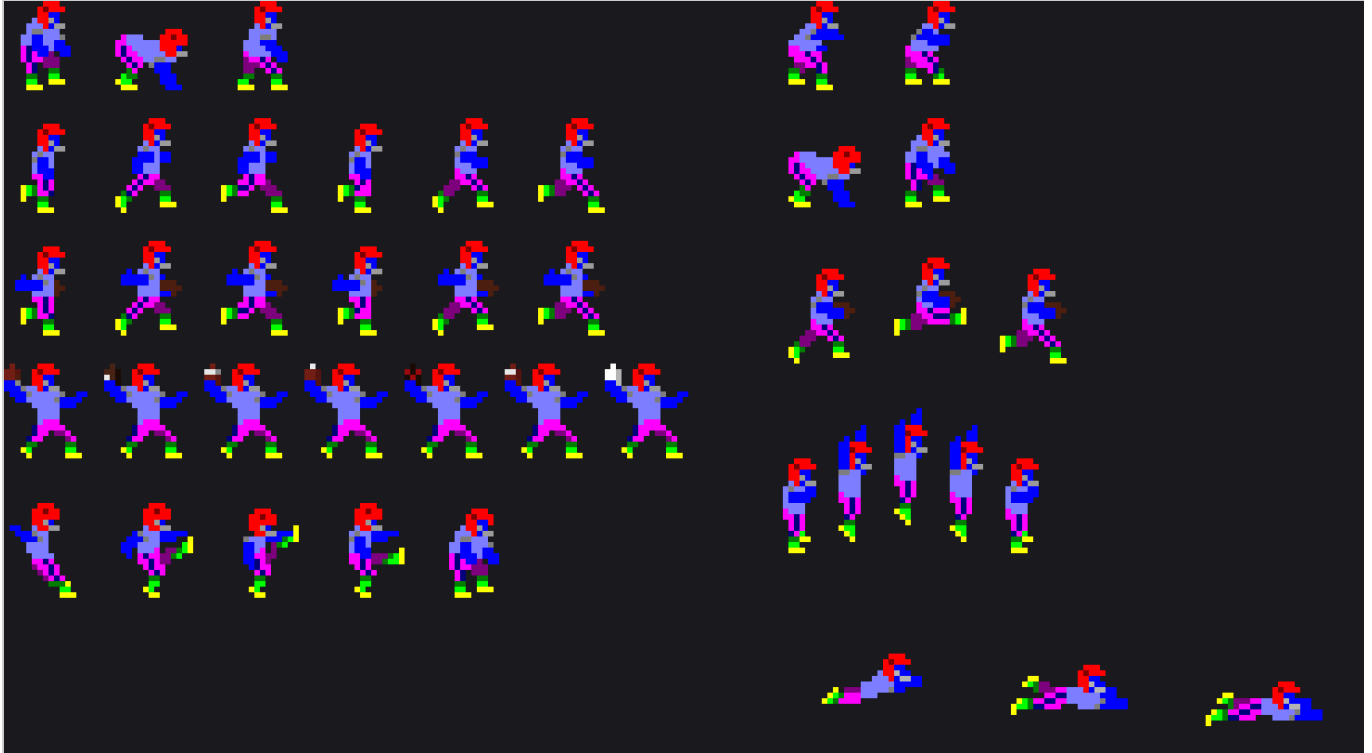
`uniforms_default.txt` ships one row per club, with eight colour slots per kit and four kits per club (home / away / alt / rush), plus three per-venue values. 32 clubs $\times$ 4 kits $\times$ 8 slots = **1,024 hand-picked colours**, in a tab-separated text file a designer can edit without opening the engine.

```
name  tla  home_helmet home_logo home_shirt home_sleeve home_pants home_stripe
      home_socks1 home_socks2 [ x4 kits ]  roof rainfall snowfall endzone
Buffalo BUF  $ffffff $003087 $003087 $c8102e $ffffff $c8102e $003087 $003087 ...
```

The last four columns are the reason two clubs' stadiums look different without any extra art: a **roof** flag, a **rainfall** and **snowfall** weighting, and an **endzone** colour. Weather is a property of the ground, not of the day.

3 The on-field figure

The single most surprising number in the build: a **Retro Bowl player is 16 × 17 pixels**, and the drawn figure inside that box is 8–13 px wide. The head is five pixels. At 270 px of screen height, a player is **6.3% of the screen**.



Every core animation at 9×, as stored. Left column: stand (3), run (6), stiff-arm (6), QB aim (7), place-kick (5). Right: wrestle (2), get-up (2), jump (3), jump-catch (5), dive (3).

The animation table

Speeds are frames per second, straight out of the manifest. A speed of — means the frames are states the code indexes into, not an animation that plays.

ACTION	CANVAS	FRAMES	FPS	CYCLE	NOTE
spr_stand	16×16	3	—	—	three <i>poses</i> , picked by state
spr_run	16×17	6	8	0.75 s	the workhorse cycle
spr_stiffarm	16×17	6	8	0.75 s	run cycle, arm out
spr_jockey	16×17	4	8	0.50 s	defender shuffle
spr_qb_walk	16×17	2	6	0.33 s	two frames is enough
spr_qb_aim	15×17	7	6	1.17 s	the most frames of any figure
spr_qb_throw	12×17	1	1	—	one frame for the throw
spr_wrestle	16×16	2	4	0.50 s	the tackle struggle — two frames
spr_getup	16×16	2	4	0.50 s	
spr_handoff	16×16	2	4	0.50 s	
spr_falling	24×24	2	3	0.67 s	bigger box — he is horizontal
spr_jump	16×24	3	8	0.38 s	taller box, same width
spr_jump_catch	7×28	5	8	0.63 s	tallest and narrowest — he is fully extended
spr_dive	32×26	3	8	0.38 s	widest box — full extension

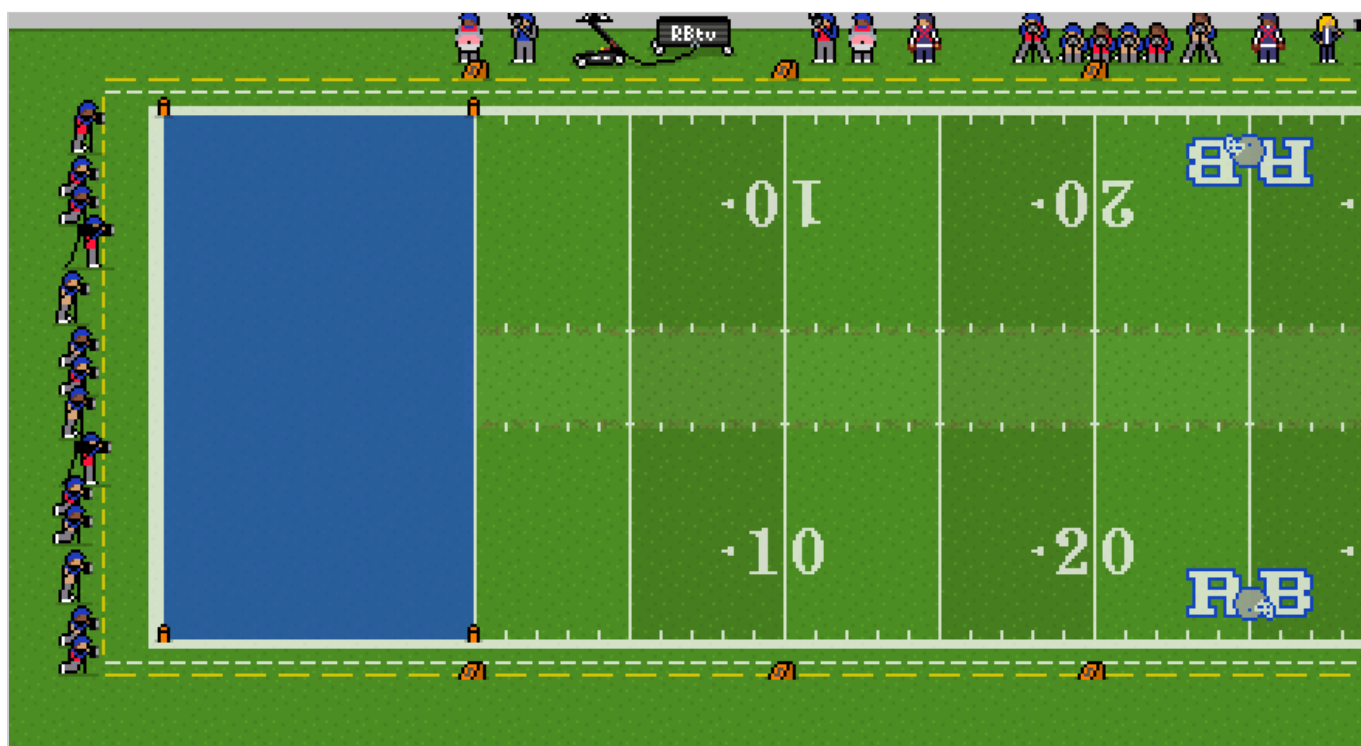
spr_spike	16×20	3	9	0.33 s	
spr_kick1 / kick2	16×20	3 / 5	8 / 9	—	punt and place-kick
spr_celebrate	16×20	2	2	1.0 s	deliberately slow

The frame budget is the headline. Thirty-two body sprites, **106 frames total**, cover every state a football player can be in. The run cycle gets six. Everything else gets two or three. A two-frame wrestle carries the most dramatic moment in the game. Nothing in Retro Bowl is animated for its own sake — frames are spent where the eye is already looking and nowhere else.

The rules the manifest reveals

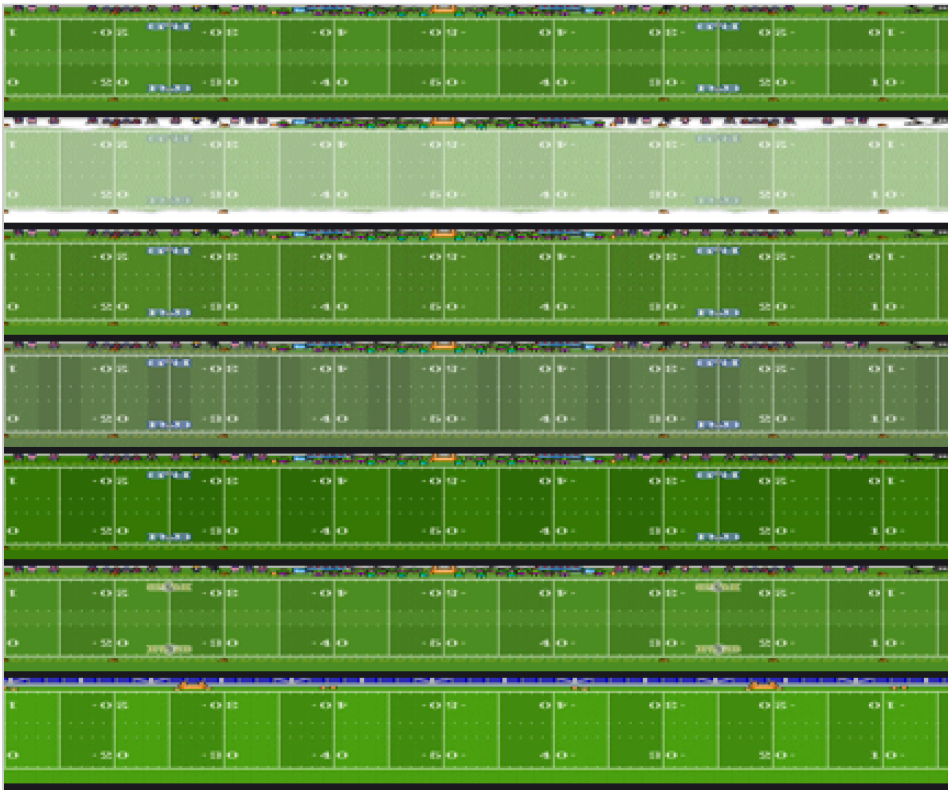
- **The canvas changes with the pose, and the origin holds the feet.** Standing poses are 16×16–17 with the origin at (7–8, 15–16) — bottom-centre. A jump grows the box *upward* to 16×24 and moves the origin to (7, 22), so the feet stay on the same ground line. A dive grows it *sideways* to 32×26. The engine never repositions the sprite; the canvas and origin do it.
- **There is one facing, mirrored.** No left-facing artwork exists in the build.
- **Ball-carrying is a parallel set, not an overlay.** `spr_run`, `spr_run_ball` and `spr_run_ball2` are three separate six-frame sets — carrying left, carrying right, and empty-handed. Three copies of the same cycle cost eighteen frames and remove every alignment problem an attached ball would have caused.
- **Loose balls are their own seven sprites.** `spr_ball1` ... `spr_ball7`, each 10×5 with four frames at 10 fps — seven rotations of a tumbling ball, plus `spr_ball` (40×20, six frames) for the ball in flight and `spr_blur` for its trail.

4 The field



The left third of `spr_field` at 2×. Yard numbers, hash marks, sideline and the near crowd are all painted into the one image. The end zone is the flat keyed block — it is recoloured per club.

One 1300 × 237 image holds the entire pitch. There is no tiling and no composition: the yard lines, the numbers, the hashes, the sideline benches and the bottom row of crowd are all baked in. Seven complete variants exist.



All seven `spr_field` frames, centre section. Reading down: standard grass, snow, a second grass dressing, worn/muddy, dark turf, a sand-patched variant, and a blue-sideline dome.

ASSET	SIZE	FRAMES	WHAT IT DOES
<code>spr_field</code>	1300×237	7	the whole pitch, seven weather/surface dressings
<code>spr_fieldfans</code>	1300×33	2	far crowd strip, two frames, 15 fps — the whole stand's animation
<code>spr_endzone</code>	100×169	7	drawn over the field; takes the club's endzone colour
<code>spr_goal_post</code>	17×90	3	three states, origin at (0,85) — the base
<code>spr_sideline_crew</code>	299×27	1	one strip of staff, repeated along the touchline
<code>spr_scoreboard</code>	208×88	1	one colour — a silhouette the game tints and writes over
<code>spr_chains</code>	32×27	2	the down markers
<code>spr_grass</code>	32×32	1	divot particles kicked up by running



Both frames of `spr_fieldfans` . A two-frame swap across a 1300-px strip is the entire crowd animation. Fans are 2-3 px each and read as a texture, not as people.

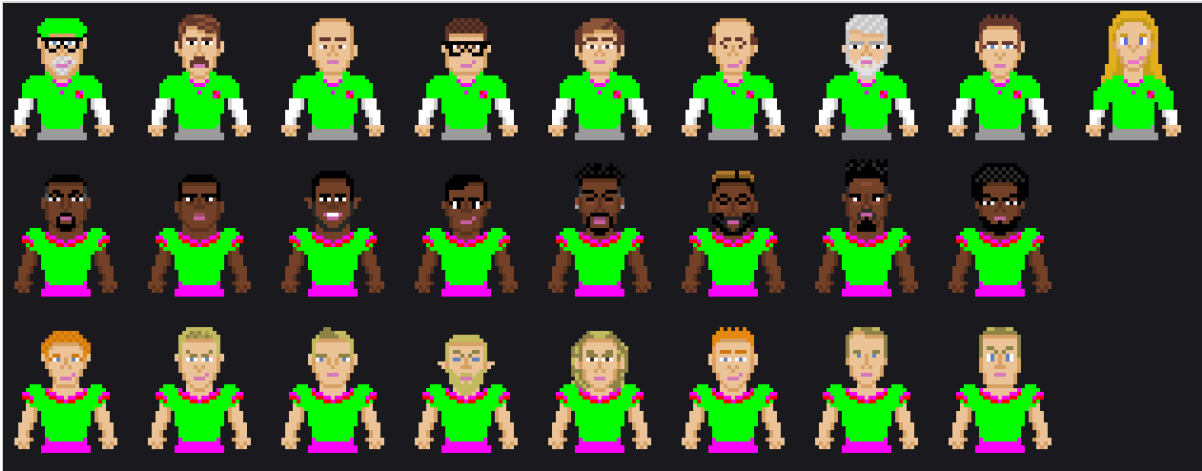
The crowd is a texture, and that is the lesson. A stand full of people costs two frames of a flat strip. Individual sideline figures — `spr_cheerleaders1..3` at 20×20 with ten frames each, `spr_coach` at 10×21 — exist only where a handful of them stand close to the camera. Everything at distance is pattern.



Sideline life and props at 6x: a cheerleader cycle, the group sprite, a coach, the chains, a goal post, a tumbling ball and the in-flight ball.

5 Portraits — where the frames actually went

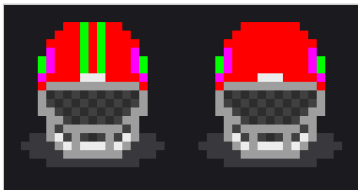
Retro Bowl spends more art on faces than on football. The portrait bank is **sixteen sprites at 32 × 36, 132 frames in total** — more frames than the entire on-field figure set, for a thing that appears on menu screens.



Three of the sixteen portrait sheets at 5×. Each is one skin tone with eight or nine head variants; the shirt is key-coloured and takes the club's colour, so one sheet dresses every team.

BANK	SHEETS	PER SHEET	TOTAL	WHAT IT IS
spr_coaches_01..04	4	9	36	staff faces
spr_dark_001..008	8	8	64	darker skin tones
spr_light_001..003	3	8	24	lighter skin tones
spr_mix_001	1	8	8	mixed

All sixteen share one canvas and one crop — head to mid-thigh, arms down, dead centre, neutral expression — so any face can appear in any frame with no layout code. The shirt and a small chest accent are keyed, which is what makes 132 drawings cover every player on every roster.



`spr_helmet`, 60×60, two frames — with and without a centre stripe. Shell, stripe and trim are three separate keys, so the team-select screen shows a real helmet in real club colours from one drawing.

6 UI furniture and type

The interface is built from fixed-size panels rather than nine-slices. Every panel exists at the exact pixel size it is used at, all of them multiples of 16 on a 480 × 270 grid: 64×32, 64×64, 64×80, 128×128, 160×128, 160×160, 176×176, 208×32, 208×64, 208×80, 208×176, 224×176, 288×160, 416×144, 448×80. Fifteen panels, one frame each. Buttons come in ten fixed widths, every one 34 px tall with three frames — up, down, disabled.

Star ratings are a single sprite with eleven frames: `spr_star_rating`, 39×7, zero to five stars in half-star steps, in six flavours (small/big, left/centre aligned, with and without a number beside it). A rating is one draw call, never five.

The five fonts, by name

FONT	TYPEFACE	SIZE	CELL	USED FOR
fnt_tecmo	Emulogic	11	15×19	the body font — monospaced, NES-era
fnt_tecmo_small	8-bit Operator+ 8	9 bold	prop. ×14	dense lists and tables
fnt_tecmo_big	8-bit Operator+ SC	16 bold	prop. ×30	headings and the scoreboard
fnt_restart	Press Start 2P	6	8×8	small caps chrome
fnt_restart_big	Press Start 2P	11	15×15	large chrome

All five are pre-rendered to the atlas with **anti-aliasing off**, which is consistent with the palette rule in §2. Press Start 2P is SIL Open Font License; Emulogic and 8-bit Operator+ are freely distributed. None of them is bespoke.

7 Totals

MEASURE	VALUE
Native canvas	480 × 270
Sprites	205
Sprite frames	632
Texture atlas pages	2 — 2048×2048 RGBA + 2048×512 indexed
Atlas entries (incl. fonts)	637
Frames on the playing figure	106 (32 sprites)
Frames on portraits	132 (16 sprites)
Key colours swapped at runtime	11
Hand-authored kit colours	1,024 (32 clubs × 4 kits × 8 slots)
Fonts	5, all third-party
Engine	GameMaker Studio 2
Pixel artist	John Savage (@pixelnfl)

8 Retro Buhurt — the locked specification

Four decisions were taken on 14 September 2026, from renders rather than opinion. This section is what an artist works from. Everything before it is the evidence.

Locked.

- **The art grid is 480×270 , drawn at $\times 2$.** The engine keeps its 960×540 coordinate space; every asset is authored at half that and the renderer doubles it. One art pixel is two screen pixels. This is Retro Bowl's own trick — its match room is a 2600-wide room holding 1300-wide art at $\times 2$ — and it is the single decision that makes this game look 8-bit rather than merely low-detail.
- **The list is 1.70 : 1.** 612 $\times$ 360 on screen, **306 $\times$ 180 authored.** Chosen from a four-way render against 1.90 (the old value), 1.50 and 1.40.
- **The list sits in a venue, not a void.** Far stand and fence above, side stands with the bench fighters at the rail, near rail and a row of crowd heads below. The old 106-px club banners and the 84-px card strip are cut to pay for it.
- **Fighters are sprites, recoloured by a key-colour shader** (§2), not engine primitives.

Still open: whether the menus and HUD move onto the coarse grid too, or stay crisp at 960×540 . That one is waiting on a render and does not block any of the art below.



The locked framing at 1.70 : 1. The floor and the fighters are the real Godot render at the new geometry; the stands, rails and crowd are blocked in to show the budget, not drawn as art. The HUD hint line and the two corner buttons still sit at their old anchors.

The vertical budget

270 authored rows, every one accounted for. Authored on the left, what it becomes on screen on the right.

BAND	AUTHORED	ROWS	ON SCREEN
HUD — clock, formation, calls	y 0-22	22	0-44
Far stand — crowd, 2 frames	y 22-40	18	44-80
Far fence / barrier	y 40-43	3	80-86
The list	y 43-223	180	86-446
Near rail	y 223-226	3	446-452
Near crowd — heads only, 2 frames	y 226-240	14	453-480
Fighter card strip	y 242-268	26	484-536

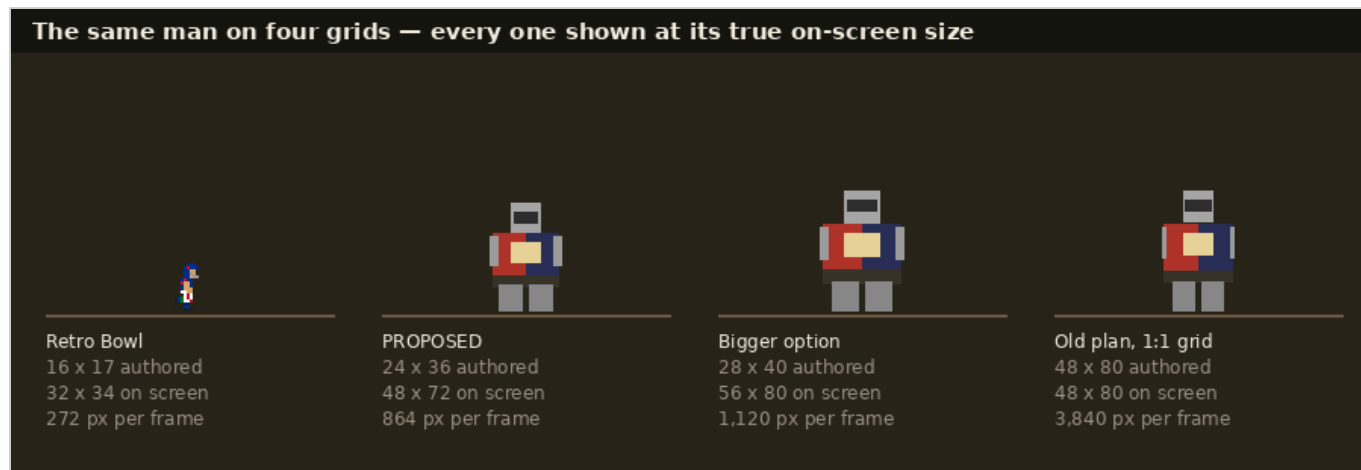
Horizontally the list runs **x 87 - 393** authored (174-822 on screen), leaving **87 authored px of side stand** on each side. The bench fighters stand at the rail inside that, crowd behind them.

The redesign does not buy list height. 184 authored rows today, 180 after. The bands added above and below eat exactly what the shrunk card strip frees. What it buys is that the 64 authored px of empty club banner down each side becomes 87 px of stand and bench, and the arena stops floating in a void. Worth knowing so nobody expects a bigger fight.

The figure

Lane pitch is 180 authored rows ÷ 5 men = **36**. So:

	AUTHORED	ON SCREEN	PIXELS PER FRAME
Retro Bowl's player	16 × 17	32 × 34	272
Retro Buhurt fighter	24 × 36	48 × 72	864
The plan this replaces	48 × 80	48 × 80	3,840



All four at their true on-screen size. The fighter blockouts are proportion studies, not art direction — they show the budget a helm, a divided surcoat and a mark have to fit into.

The fighter is barely smaller on screen and 4.4× cheaper to draw. 72 screen px tall against the old plan's 80, for 864 pixels a frame instead of 3,840. The grid decision does not shrink the man, it coarsens him. And 24 × 36 lands within four pixels of the **24 × 32** that REGISTER 05.3 measured by rendering as the size where the heraldry finally did any work — the only figure size in this project's history arrived at by looking rather than by arithmetic.

If more room to draw is wanted later there is exactly one lever: float the card strip over the list the way Retro Bowl floats its skip button. That reclaims 26 authored rows, giving a 206-row list, a 41-px lane and a **28 × 40** fighter.

The key colours

Nine slots, replacing Retro Bowl's eleven. Skin is dropped — the men are in helms — and what it frees goes to the heraldic division of the surcoat, which is what a club is actually recognised by. The three rules from §2 apply unchanged: **opaque pixels only, exact hex, never eyedropped, the shade is its own slot, not a darkening.**

SLOT	KEY	WHAT IT PAINTS
surcoat_a		#FF00FF surcoat field, primary tincture
surcoat_b		#800080 second tincture — per pale, per fess, quartered
trim		#FFFF00 edging, belt, straps
helm		#FF0000 helm cover, plume, crest
steel		#8080FF plate — lets a club run blackened or polished
steel_b		#0000FF plate shadow
leather		#00FF00 gambeson, gauntlet cuffs, boots
leather_b		#008000 leather shadow
mark		#808080 a flat chest panel where IconBank stamps the club mark

The fidelity — how 8-bit, exactly

Measured rather than asserted. Retro Bowl is **NES-sized and SNES-coloured**, and it keeps the two apart rigidly.

	NES	SNES	RETRO BOWL
Canvas	256 × 240	256 × 224	480 × 270
Colours per sprite	3 + transparent	up to 16	11-13
Figure size	8×8 / 8×16 tiles	—	16 × 17 (≈ 2 × 2 tiles)

Per-frame counts from the atlas: `spr_stand` 11-12, `spr_run` 12-13, `spr_wrestle` 12, `spr_dive` 11-12. The one outlier, `spr_qb_aim` at 14-16, is carrying a ball and a throwing arm.

It drops hard for anything that is not a person — helmet 7, cheerleader 8, coach 8, chains 6, goal post 4-6, loose ball **2** — and UI panels are **one colour**: `spr_box_208x80`, `spr_scoreboard` and `spr_star_rating` are single-colour silhouettes the game tints and writes over. The premium is spent entirely on **backgrounds**, where `spr_field` runs 41-212 colours a frame.

Binary alpha on every figure. Across all 632 frames, 31 sprites contain partial alpha and every one of them is UI or an effect — HUD panels, control hints, the clock, rain, motion blur. Of the eighteen player sprite families, **not one pixel is anything other than fully transparent or fully opaque.** That is not a style choice: the colour-replace shader tests `pixel.a == 1.0`, so a soft edge on a fighter renders as raw magenta.

What carries the 8-bit read, and all of it transfers: the figure at roughly two tiles, flat fills with no outline and no gradient, binary alpha, two to three frames per action with six for the one cycle that matters, and arcade-era type. Dithering only where there is room for it — they checker the 60 × 60 helmet's facemask and dither nothing on a 16 × 17 man.

The budget for a 24 × 36 fighter: twelve colours, hard ceiling fourteen

Nine are the key slots above. Three are fixed and never swapped:

#	COLOUR	JOB
---	--------	-----

1-9	the nine keys	steel, steel_b, helm, surcoat_a, surcoat_b, trim, leather, leather_b, mark
10	black	visor slot, the deepest separation between overlapping men
11	near-black	one step up — under the arms, behind the belt
12	white	a single specular hit on the plate. One.

Why NES-strict would break this game specifically. Armour reads as metal through value steps. At three colours the harness eats the whole budget and the surcoat gets nothing — and the surcoat is the only thing that tells two grey men apart. REGISTER 05.3 measured that failure once already from the other direction, when the figure was too small for the heraldry to work. **The 8-bit is carried by size, flatness and frame count, not by a colour cap.**

The crowd

Retro Bowl's crowd strip is **21 colours, and the two most common are #262626 and #3B3B3B** — a dark mass with club-coloured specks and a few skin tones, not a rainbow. Two of the twenty-one are shader keys, so **the crowd wears the home club's kit**. Copy that exactly: it is why their stands recede instead of competing with the play.

A crowd figure is 3×4 authored pixels — a two-pixel head and a three-pixel body. Two frames per stand is the whole animation.

The work order

Every canvas below is the authored size. The renderer doubles it. Nothing is drawn with anti-aliasing, a soft edge, or a partial-alpha pixel.

ASSET	CANVAS	FRAMES	NOTE
The fighter — one facing, mirrored. Origin bottom-centre, feet on the ground line.			
idle / guard	24 × 36	2-3	
walk	24 × 36	4	
charge	24 × 36	6	the one cycle worth six frames
strike	24 × 36	3	
clinch	32 × 36	2	wider box — two men, like their 32×26 dive
takedown / thrown	40 × 36	3	grows sideways, origin holds the feet
downed	40 × 22	1-2	
get up	24 × 36	2	
~25 frames total. At 864 px a frame that is about 21,600 pixels — three quarters of Retro Bowl's entire on-field cast.			
The venue			
List surface	306 × 180	2-4	whole variants (dry, wet, snow, indoor) — never a tint
Far stand	480 × 18	2	full width, two-frame swap
Side stand	87 × 180	2	one drawing, mirrored
Near crowd, heads	480 × 14	2	
Rail / fence	480 × 3	1	tiles horizontally
Everything else, halved from the old brief			
Arena grounds	260 × 150	6	was 520×300
Venue splashes (away, neutral)	480 × 270	2	was 960×540
Portrait harnesses	32 × 32	4	was 64×64 — now Retro Bowl's own portrait size
Portrait heads	32 × 32	16	same shared canvas, gorget line at y 19

The two-layer portrait scheme survives the halving and should not be given up: four harnesses × sixteen heads is **64 men from 20 drawings**, against the 132 separate frames Retro Bowl draws for the same job.

What has to change in the code before any of this lands

1. `Tuning.LIST_H` 570 → **510**. This is the one constant that invalidates the whole melee suite; everything is re-measured after it moves, once, deliberately.
2. `melee_scene.LIST_SCALE` 1.23 → **1.20**, `LIST_ORIGIN.y` 58 → **86**.
3. Banners out; far / side / near stand bands in. Card strip `CARD_H` 84 → 52, `STRIP_Y` 436 → 484. The two corner buttons need re-anchoring.
4. A fighter-sprite slot in `ArtBank.SLOTS` — there is none today, and `test_arena.gd` asserts the table and `ART.md` agree, so art can be finished with nowhere to land.
5. **Delete the greyscale rule** from `ART.md` and `ArtBank`. It exists to serve the mood tint, which was dropped on 14 September. Retro Bowl ships seven coloured field variants and tints nothing; variants is the answer.
6. Halve every `size` in `ArtBank.SLOTS` and draw at ×2.
7. A `shader_type canvas_item` port of §2 — nine equality tests, opaque pixels only, ±0.009.

9 Full sprite index

All 205 sprites, all 632 frames, straight from the manifest. Canvas is the authored PNG size; origin is the draw anchor in pixels from the top-left; fps is the manifest playback speed, and — means the frames are indexed states rather than an animation.

Body — the on-field figure 32 sprites · 106 frames

SPRITE	CANVAS	FRAMES	FPS	ORIGIN
spr_stand	16×16	3	—	8,15
spr_stand_ball	16×16	3	—	7,15
spr_stand_ball2	16×16	3	—	7,15
spr_stand_highlight	16×24	3	—	8,22
spr_run	16×17	6	8	7,16
spr_run_ball	16×16	6	8	7,15
spr_run_ball2	16×16	6	8	7,15
spr_stiffarm	16×17	6	8	7,15
spr_stiffarm2	16×17	6	8	7,15
spr_wrestle	16×16	2	4	9,15
spr_getup	16×16	2	4	7,15
spr_getup_ball	16×16	2	4	7,15
spr_getup_ball2	16×16	2	4	7,15
spr_jump	16×24	3	8	7,22
spr_jump2	16×24	3	8	7,22
spr_jump_catch	7×28	5	8	3,27
spr_jump_catch_ball	7×28	5	8	3,27
spr_jump_catch_ball2	7×28	5	8	3,27
spr_dive	32×26	3	8	16,25
spr_dive_ball	32×26	3	8	16,25
spr_dive_ball2	32×26	3	8	8,17
spr_falling	24×24	2	3	12,22
spr_falling_ball	24×24	2	3	12,22
spr_falling_ball2	24×24	2	3	12,22
spr_spike	16×20	3	9	8,19
spr_spike2	16×20	3	9	8,19
spr_handoff	16×16	2	4	8,15
spr_celebrate	16×20	2	2	8,19
spr_dummy	16×16	3	—	7,13
spr_jockey	16×17	4	8	7,15
spr_place	16×13	2	—	6,11
spr_highlight	16×16	1	15	8,7

Quarterback and kicker 9 sprites · 22 frames

SPRITE	CANVAS	FRAMES	FPS	ORIGIN
spr_qb_walk	16×17	2	6	8,16
spr_qb_aim	15×17	7	6	7,16
spr_qb_aim_old	15×17	1	6	7,16
spr_qb_throw	12×17	1	1	6,16
spr_kick1	16×20	3	8	8,17
spr_kick2	16×20	5	9	8,17
spr_kickpoint	25×7	1	15	6,3
spr_aim	7×7	1	15	2,3
spr_route_pointR	16×16	1	15	8,7

The ball 10 sprites · 36 frames

SPRITE	CANVAS	FRAMES	FPS	ORIGIN
spr_ball	40×20	6	—	20,10
spr_ball1	10×5	4	10	5,2
spr_ball2	10×5	4	10	5,2
spr_ball3	10×5	4	10	5,2
spr_ball4	10×5	4	10	5,2
spr_ball5	10×5	4	10	5,2
spr_ball6	10×5	4	10	5,2
spr_ball7	10×5	4	10	5,2
spr_blur	10×5	1	7	5,1
spr_bouncespot	16×16	1	15	4,3

Field and venue 12 sprites · 29 frames

SPRITE	CANVAS	FRAMES	FPS	ORIGIN
spr_field	1300×237	7	—	0,0
spr_fieldfans	1300×33	2	15	0,0
spr_endzone	100×169	7	15	0,0
spr_goal_post	17×90	3	—	0,85
spr_goal_stand	10×37	2	—	0,32
spr_goal_shadow	32×35	1	—	3,17
spr_chains	32×27	2	15	16,26
spr_sideline_crew	299×27	1	15	0,0
spr_grass	32×32	1	15	0,0
spr_cone	7×4	1	15	3,2
spr_scoreboard	208×88	1	15	0,0
spr_matchline_dot	32×16	1	15	16,8

Sideline life 10 sprites · 39 frames

SPRITE	CANVAS	FRAMES	FPS	ORIGIN
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spr_cheerleaders	100×25	2	2	50,24
spr_cheerleaders1	20×20	10	4	10,20
spr_cheerleaders2	20×20	10	4	10,20
spr_cheerleaders3	20×20	10	4	10,20
spr_coach	10×21	1	15	5,19
spr_fans	12×14	1	15	6,7
spr_team_celebration	100×40	1	15	0,0
spr_press_conf	100×40	1	15	0,0
spr_commentary_bg	16×16	2	—	0,0
spr_spangle	23×7	1	15	11,4

Weather and effects 7 sprites · 15 frames

SPRITE	CANVAS	FRAMES	FPS	ORIGIN
spr_rain	2×8	2	8	0,7
spr_snow	3×3	1	9	1,1
spr_splash	13×12	5	3	6,9
spr_scanlines	64×60	1	15	0,0
spr_square	16×16	1	15	0,0
spr_box	4×4	4	15	0,0
spr_wind	22×21	1	15	9,10

Portraits and the face bank 19 sprites · 137 frames

SPRITE	CANVAS	FRAMES	FPS	ORIGIN
spr_coaches_01	32×36	9	15	0,0
spr_coaches_02	32×36	9	15	0,0
spr_coaches_03	32×36	9	15	0,0
spr_coaches_04	32×36	9	15	0,0
spr_dark_001	32×36	8	15	0,0
spr_dark_002	32×36	8	15	0,0
spr_dark_003	32×36	8	15	0,0
spr_dark_004	32×36	8	15	0,0
spr_dark_005	32×36	8	15	0,0
spr_dark_006	32×36	8	15	0,0
spr_dark_007	32×36	8	15	0,0
spr_dark_008	32×36	8	15	0,0
spr_light_001	32×36	8	15	0,0
spr_light_002	32×36	8	15	0,0
spr_light_003	32×36	8	15	0,0
spr_mix_001	32×36	8	15	0,0
spr_helmet	60×60	2	—	30,30
spr_jacket	32×32	1	15	0,0
spr_player_edit	15×20	2	—	7,10

Ratings, meters and trophies 31 sprites · 122 frames

SPRITE	CANVAS	FRAMES	FPS	ORIGIN
spr_star	7×7	1	15	3,4
spr_star_b	9×9	1	15	4,4
spr_star_s	5×4	1	15	2,2
spr_star_rating	39×7	11	15	0,3
spr_star_rating_b	41×9	11	15	0,4
spr_star_rating_bn	41×9	11	15	19,3
spr_star_rating_c	39×7	11	15	19,3
spr_star_rating_cb	41×9	11	15	20,4
spr_star_rating_cbn	41×9	11	15	20,4
spr_attitude	11×11	7	15	5,5
spr_attitude_s	5×5	7	15	2,2
spr_condition	11×11	4	15	5,5
spr_morale	9×9	1	15	4,4
spr_workrate	96×32	3	—	0,0
spr_meter	192×20	1	15	0,0
spr_xp	14×7	1	15	7,3
spr_credit	9×9	6	—	4,4
spr_credit2	14×9	1	—	6,4
spr_heart	9×9	1	15	4,5
spr_rookie	11×11	1	—	5,5
spr_trade	11×11	1	15	5,5
spr_medals	7×12	4	15	0,0
spr_medals_c	7×12	4	15	3,5
spr_sb_ring	11×11	1	15	0,0
spr_sb_ring_c	11×11	1	15	5,5
spr_cup	15×15	1	15	7,7
spr_trophy	32×32	1	15	15,15
spr_hof	32×32	2	—	0,0
spr_hof_btn	32×32	2	—	16,16
spr_hof_c	32×32	2	—	16,16
spr_hof_s	7×8	1	15	3,0

Panels, buttons and chrome 75 sprites · 126 frames

SPRITE	CANVAS	FRAMES	FPS	ORIGIN
spr_adRG	480×270	1	30	0,0
spr_app_apple	110×32	1	15	0,0
spr_app_discord	110×32	1	—	0,0
spr_app_google	110×32	1	15	0,0
spr_app_reddit	110×32	1	15	0,0
spr_app_twitter	110×32	1	—	0,0

spr_arrowL	32×32	1	15	16,16
spr_arrowL2	8×15	1	15	7,7
spr_arrowR	32×32	1	15	16,16
spr_arrowR2	8×15	1	15	2,7
spr_arrowR3	6×13	1	15	2,6
spr_arrowU	32×32	1	15	16,16
spr_audible	96×32	5	—	0,0
spr_box_128x128	128×128	1	15	0,0
spr_box_160x128	160×128	1	15	0,0
spr_box_160x160	160×160	1	15	0,0
spr_box_176x176	176×176	1	15	0,0
spr_box_208x176	208×176	1	15	0,0
spr_box_208x32	208×32	1	15	0,0
spr_box_208x64	208×64	1	15	0,0
spr_box_208x80	208×80	1	15	0,0
spr_box_224x176	224×176	1	15	0,0
spr_box_288x160	288×160	1	15	0,0
spr_box_416x144	416×144	1	15	0,0
spr_box_416x144_fill	416×144	1	15	0,0
spr_box_448x80	448×80	1	15	0,0
spr_box_64x32	64×32	1	15	0,0
spr_box_64x41	64×41	1	15	0,0
spr_box_64x64	64×64	1	15	0,0
spr_box_64x80	64×80	1	15	0,0
spr_button_104	106×34	3	—	0,0
spr_button_32	33×34	3	—	0,0
spr_button_32_9	33×34	3	—	0,0
spr_button_40	41×34	3	—	0,0
spr_button_50	54×34	3	—	0,0
spr_button_64	68×34	3	—	0,0
spr_button_74	74×34	3	—	0,0
spr_button_74s	74×26	3	15	0,0
spr_button_90	90×34	3	—	0,0
spr_button_thin	41×18	3	—	0,0
spr_checkbox	16×16	2	—	0,0
spr_checkbox012	16×16	3	—	0,0
spr_clock	96×32	3	15	0,0
spr_clock2	96×32	2	15	0,0
spr_col_picker	391×262	1	15	195,131
spr_connect	7×7	1	15	3,3
spr_controls	140×59	1	15	70,29
spr_controls_handoff	140×59	1	15	70,29

spr_controls_kick	140×59	1	15	70,29
spr_controls_run	140×59	1	15	70,29
spr_controls_runb	140×59	1	15	70,29
spr_controls_scramble	140×59	1	15	70,29
spr_controls_throw	140×59	1	15	70,29
spr_cross	32×32	1	15	15,16
spr_details	32×32	1	15	16,16
spr_edit	14×14	1	15	7,7
spr_fieldgoalbtn	96×32	1	—	0,0
spr_firstdown	20×20	7	4	10,19
spr_gamepad_any	20×20	4	15	12,7
spr_gamepad_xbox1	20×20	4	15	12,7
spr_home	32×32	1	15	15,15
spr_info	16×16	1	15	0,0
spr_msg_bg	64×64	1	15	0,0
spr_pointer	9×12	2	—	4,6
spr_profile	32×32	1	15	16,16
spr_spanner	32×32	1	15	15,15
spr_store	15×15	1	15	7,7
spr_textbox48	48×16	1	15	24,8
spr_textbox64	64×16	1	15	32,8
spr_tick	32×32	1	15	16,16
spr_training	16×23	1	15	8,11
spr_uniform_select	46×2	5	15	23,1
spr_uniform_select2	36×2	2	15	18,1
spr_uniform_select4	36×2	4	—	7,10
spr_watermark	960×540	1	15	480,270

10 Method and trust

The build was obtained as a public GameMaker HTML5 export and parsed, not played and eyeballed. The asset manifest inside `RetroBowl.js` carries, per sprite: name, authored width and height, origin, bounding box, playback speed, and the index of every frame into a 637-entry texture table that gives each frame's page, position, crop and untrimmed size. Sprite sizes were cross-checked against the atlas crops and against sprites whose names encode their own dimensions (`spr_box_128x128` is 128×128 in both places). Room and camera sizes come from the room definitions; the shader is quoted verbatim; the key colours are the literal uniform values set in the calling code; fonts are the typeface names recorded in the font definitions.

Two things in this document are not measurement. The **reading of intent** — why frames are spent where they are, why the crowd is a texture — is analysis. The **Retro Buhurt proposals** in §8 are recommendations, with the arithmetic behind each one shown so it can be checked rather than taken. The Retro Buhurt figures themselves are read out of the current source.

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